Genetics: an overview

5A



Genetics is a branch of biology. It is the study of inheritance — the passing on of characteristics from one generation to the next. The term 'genetics' was first used by William Bateson (a British biologist) in 1906.

Although theories of inheritance, or heredity, were put forward at least as early as the 5th and 4th centuries BC, the study of genetics only truly began in the 18th and 19th centuries. Observations were made of how specific characteristics of plants and animals were passed on from one generation to the next, to provide a rational basis for the improvement of crop plants and livestock.

The most significant breakthrough in genetics was made by the Austrian Monk Gregor Mendel (1822–84). He observed specific features of the pea plant and counted the number of individuals in which each characteristic appeared through several generations. By concentrating on just a few features and determining what proportion of each generation received them, he was able to demonstrate specific patterns of inheritance. These patterns have been shown to be applicable to most genetic systems.

There are two ways in which living things reproduce themselves — sexually and asexually. This sheet deals with sexual reproduction, which leads to genetic variation.

All animals and plants which reproduce sexually start with a female sex cell (egg or ovum) which is entered and fertilised by a male sex cell (in animals this is sperm; in flowering plants it is pollen). Sex cells are called *gametes*.

The fertilised egg (zygote) grows into a new individual by a series of cell divisions. The new individual is of the same species and has the *characteristics* of its parent cells. The 'recipe' for this is carried in the parent gametes. The structures which cause this *inheritance* of these characteristics are called *chromosomes*. Each chromosome is made up of *genes* — the 'coded' instructions for the appearance and constituents of an organism.

> **Chromosomes:** thread-like structures containing the hereditary information found in the nucleus of the cell. They consist of DNA. They are paired — one from each parent. Each chromosome contains information for a number of characteristics, such as hair colour, eye colour, shape of ear, etc.

Chromosomes are present at all times in the nuclei of all cells. Each chromosome is made of a single molecule of *DNA* (deoxyribonucleic acid) which consists of a chain of many connected genes. Every species has its own number of chromosomes per cell, arranged in pairs. Humans have 46 chromosomes in body cells and 23 in sex cells.

Note that sex cells have half the number of chromosomes that are present in other cells. Gametes are produced by a type of cell division called *meiosis*, in which 'daughter' cells receive exactly half the number of chromosomes as the original nucleus. When non-sex cells (somatic cells) divide for growth or repair, the daughter cells receive the same number of chromosomes as the original by a type of cell division called *mitosis*.

When sex cells come together at fertilisation the 'correct' number of chromosomes is restored. The new individual gets half of its chromosomes from each parent sex cell, so the new individual's cell has the total number of chromosomes of both parent cells combined.

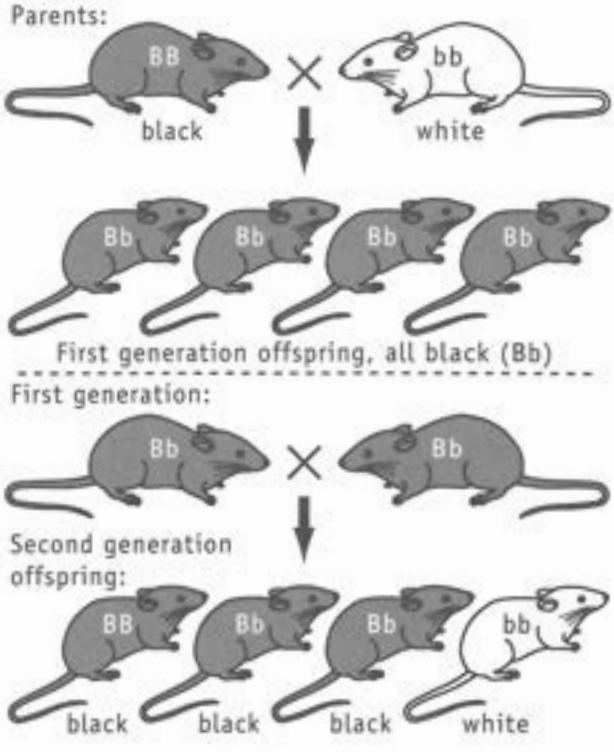
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Genetics: an overview

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Genes are the basic unit of inheritance and relate to specific characteristics (traits) of organisms. In humans, for example, these could be blood group, hair colour, etc. These genes (with the exception of the sex chromosomes) are paired. These paired genes (one from each parent) control the same characteristic and may give identical instructions. However their instructions may also be different, in which case the instructions from the *dominant* gene will 'mask out' those from the *recessive* gene. Such non-identical genes are called *alleles*.



On this sheet, we look at an example of single-factor inheritance that follows the laws established by Mendel — the colouration of rats. In this case the allele for black is dominant over the allele for white.

If we call the allele for white b and the allele for black B, it implies that the allele with the capital letter is dominant.

If a homozygous black rat (with identical alleles BB) mates with a homozygous white rat (bb), all the offspring will be heterozygous black (Bb).

If these offspring are then mated, they give a mixture of black rats and white rats. The ratio of black rats to white rats will be 3 : 1.

1. Define 'genetics'.
2. Who could be said to have been the 'father' of modern genetics? What organisms did he study?
3. Which type of reproduction leads to genetic variation?
4. What is the general name for a sex cell?
5. What do we call a fertilised egg?
6. State the number of chromosomes present in human body cells.
7. State the number of chromosomes present in human sex cells.
8. List the difference between 'mitosis' and 'meiosis'.
9. Briefly explain what is meant by 'allele'.
10. Briefly explain what is meant by 'homozygous'.
11. Briefly explain what is meant by 'heterozygous'.
12. 'Sexual reproduction gives offspring that are identical to the parents.' True or false?
13. 'Chromosomes are found in the nucleus of a cell.' True or false?
14. 'Genes are the basic units of inheritance.' True or false?
15. 'The term "dominant" refers to the characteristic that can always be seen in the offspring, for example brown eye colour is dominant over blue eye colour.' True or false?

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